

Benchmarking Citation Fabrication Rates Across Large Language Models and Agentic Search Tools

A Comparative Review of Fabrication Rates, Detection Methods, and Mitigation Strategies

Abstract

Citation fabrication — the generation of bibliographic references, case citations, or source attributions that do not correspond to real, verifiable works — has emerged as one of the most consequential and measurable forms of hallucination in large language models (LLMs) and the agentic search systems built on top of them. This paper synthesizes empirical benchmarking work published between 2023 and 2026 that quantifies citation fabrication rates across general-purpose chatbots (ChatGPT, Bard/Gemini, Claude, Grok, DeepSeek), retrieval-augmented generation (RAG) pipelines, commercial AI legal research tools, and agentic "deep research" systems that autonomously browse and cite web sources. Reported fabrication rates range from roughly 14% to 95% depending on model, domain, and prompting condition, with GPT-3.5-era models fabricating citations in the range of 55%, GPT-4-class models improving to roughly 18%, and RAG-grounded commercial products still hallucinating in 17–33% of outputs despite explicit "hallucination-free" marketing claims. Agentic search tools evaluated by the Columbia Journalism Review's Tow Center for Digital Journalism produced incorrect source attributions in more than 60% of tested queries, and a large-scale audit of deep research agents found that such agents generate more citations per query than search-augmented chatbots yet hallucinate URLs at higher per-citation rates. We organize this literature around a taxonomy of fabrication types (non-existent works, author-title mismatches, metadata corruption, and unsupported attribution), review current detection and mitigation approaches (retrieval grounding, sampling-based consistency checking, and post-hoc verification pipelines), and identify open methodological challenges, including the absence of standardized fabrication metrics, inconsistent human-verification protocols, and the difficulty of benchmarking closed commercial agentic systems. We conclude with a research agenda emphasizing standardized benchmarks, longitudinal tracking as models are updated, and verification infrastructure integrated into scholarly and legal publishing workflows.

Keywords: citation fabrication; hallucination; large language models; retrieval-augmented generation; agentic search; benchmarking; scholarly integrity; AI-assisted writing; deep research agents; legal AI

1. Introduction

The rapid integration of large language models into academic writing, legal research, journalism, and web search has made citation generation one of the most visible and consequential surfaces on which LLM reliability is tested. Unlike many forms of hallucination, which require domain expertise to detect, fabricated citations are in principle externally auditable: a claimed bibliographic reference either resolves to a real, matching work in an authoritative database or it does not. This tractability has made citation fabrication a popular and comparatively rigorous benchmarking target, producing a body of empirical work that spans general-purpose chatbots, retrieval-augmented systems marketed as hallucination-

resistant, and increasingly autonomous "agentic" search and deep-research tools that browse the live web and synthesize cited reports on a user's behalf.

The stakes of this problem are not merely academic. Fabricated citations generated by ChatGPT have been implicated in courtroom sanctions after attorneys submitted briefs containing non-existent case law, in the withdrawal of peer-reviewed papers found to cite invented sources, and in the propagation of confidently stated but non-existent scientific evidence in mental-health and biomedical writing (Rao, Wong, & Callison-Burch, 2026; Dahl, Magesh, Suzgun, & Ho, 2024; Linardon, Jarman, McClure, Anderson, Liu, & Messer, 2025). As LLMs are embedded ever more deeply into research assistance, legal practice, and "agentic" search products that autonomously retrieve and cite web content, understanding how frequently these systems fabricate citations — and under what conditions — has become an urgent question for scholarly integrity, legal practice, journalism, and public trust in AI-mediated information.

This paper reviews and synthesizes the empirical literature on citation fabrication benchmarking, with three specific aims. First, we characterize how citation fabrication has been operationalized and measured across studies, since inconsistent definitions and verification protocols make cross-study comparison difficult. Second, we compare reported fabrication rates across model families, domains, and system types — general-purpose LLMs, retrieval-augmented generation (RAG) systems, agentic deep-research tools, and generative search engines — to establish where the field currently stands. Third, we review current detection and mitigation approaches and identify the research gaps that remain, including the absence of standardized benchmarks, limited longitudinal tracking as models are updated, and the difficulty of auditing closed commercial agentic systems whose retrieval and reasoning pipelines are not publicly documented.

2. Background, Related Work, and Literature Review

2.1 Hallucination as a Foundational Concept

Citation fabrication is a specific instance of the broader phenomenon of hallucination in natural language generation, defined by Ji et al. (2023) as generated content that is nonsensical or unfaithful to the provided source material or to verifiable world knowledge. Ji et al.'s survey distinguishes intrinsic hallucination, in which output content directly contradicts an available source, from extrinsic hallucination, in which content cannot be verified against the source at all — a category into which most fabricated citations fall, since the model typically has no source document constraining its output and instead generates plausible-looking bibliographic metadata from parametric memory or pattern completion. This distinction matters because it implies that citation fabrication is not primarily a factual-recall failure but a byproduct of how autoregressive language models generate structurally plausible text: a well-formed citation (author names, a plausible title, a real journal name, a numerically plausible year and volume) can be highly fluent while referring to nothing that exists.

2.2 Early Case Studies and the First Systematic Benchmarks

The earliest documented reports of LLM citation fabrication were case studies rather than benchmarks. Alkaissi and McFarlane (2023) reported that when ChatGPT was asked to generate references for two

biomedical case reports, all five citations returned for one topic had non-existent titles, and the PubMed identifiers (PMIDs) attached to them pointed to unrelated, real papers — a pattern in which fabricated content is deliberately dressed in genuine-looking identifiers. When the same prompt was repeated, ChatGPT returned different fabricated titles with different (still incorrect) PMIDs, illustrating the stochastic and non-reproducible nature of the fabrication.

The first large-scale, systematic study was conducted by Walters and Wilder (2023), who prompted GPT-3.5 and GPT-4 to generate 84 short literature reviews across 42 multidisciplinary topics and manually verified all 636 resulting citations against library databases. They found that 55% of GPT-3.5's citations were entirely fabricated compared with 18% for GPT-4, and that even among non-fabricated (real) citations, 43% (GPT-3.5) and 24% (GPT-4) contained substantive errors such as incorrect volume, issue, or page numbers. Notably, fabricated citations almost always used real, existing journal names combined with invented article titles, and hyperlinks were more frequent in fabricated citations (18%) than in real ones (5%) — evidence that superficial markers of credibility (a working-looking link, a legitimate venue name) do not reliably signal citation validity.

Mugaanyi, Cai, Cheng, Lu, and Huang (2024) extended this line of work by comparing GPT-3.5's citation behavior across natural-science and humanities topics, finding no significant difference in the rate of entirely non-existent citations between disciplines (approximately 23–27%) but a substantially higher rate of hallucinated digital object identifiers (DOIs) in humanities citations (89.4%) than in natural-science citations (61.8% in one measurement, and as low as 29.1% in a related analysis), attributing the disparity to the model's tendency to interpolate plausible-looking identifiers when it cannot retrieve genuine metadata rather than acknowledging the absence of a real DOI. Building on this, subsequent replications reported by Naser (2026) similarly found citation hallucination to vary between 11.4% and 56.8% across ten commercially deployed models and four academic domains, and to be strongly shaped by prompt framing — with no model observed to fabricate citations spontaneously in the absence of a citation-eliciting prompt, suggesting the behavior is prompt-induced rather than an unconditional model tendency.

2.3 Domain-Specific Findings: Medicine, Mental Health, and Law

In the medical and mental-health domains, fabrication rates have shown pronounced sensitivity to topic familiarity. Linardon et al. (2025) prompted GPT-4o to generate literature reviews on mental-health disorders varying in public visibility and found that 19.9% of all citations across six simulated reviews were entirely fabricated, but that this rate ranged from approximately 6% for a highly familiar disorder (major depressive disorder) to 28–29% for a less visible, less-researched disorder (body dysmorphic disorder), indicating that citation reliability is not a fixed model property but interacts with the density and public prominence of a topic's underlying training data. A related score-based approach was proposed by Aljamaan et al. (2024), who developed a "reference hallucination score" for medical AI chatbots to standardize how fabrication severity is quantified across systems, an early move toward the kind of standardized metric this literature still largely lacks.

Legal applications reveal comparably severe, and in some respects worse, outcomes because legal citation accuracy carries direct professional and due-process consequences. Dahl, Magesh, Suzgun, and Ho (2024) constructed an original suite of legal queries and found that general-purpose LLMs hallucinate between 58% (GPT-4) and 88% (Llama 2) of the time when asked specific, verifiable questions about federal court cases, and that models frequently fail to correct a user's mistaken legal premise rather than flagging the error. Critically, Magesh, Surani, Dahl, Suzgun, Manning, and Ho (2025) subsequently conducted the first pre-registered empirical evaluation of commercial AI legal research tools — including Lexis+ AI and Westlaw AI-Assisted Research — that explicitly market retrieval-augmented generation as eliminating or avoiding hallucination. They found that, although hallucination rates were meaningfully lower than for general-purpose GPT-4 use, these RAG-grounded commercial products still hallucinated in 17% to 33% of tested queries, directly contradicting vendor claims of "hallucination-free" citations and demonstrating that retrieval grounding substantially reduces but does not eliminate fabrication.

2.4 Agentic Search and Deep Research Tools

A distinct and increasingly important line of work examines agentic systems: LLM-driven tools that autonomously issue search queries, browse retrieved pages, and synthesize cited answers or reports, including consumer-facing "AI search engines" (ChatGPT Search, Perplexity, Gemini, Copilot, Grok Search) and "deep research" agents offered by major model providers. The Tow Center for Digital Journalism at Columbia University conducted the most widely cited evaluation of this class of tool, testing eight generative AI search engines on 1,600 queries that asked each system to identify the source publication, date, and URL for a supplied direct quotation from a news article (Jaźwińska & Chandrasekar, 2025). Across all eight tools, incorrect citations were returned more than 60% of the time; the best-performing tool, Perplexity, still answered incorrectly in 37% of cases, while the worst-performing tool exceeded 90% error. The study also found that these tools were poor at declining to answer when they lacked sufficient information, frequently fabricating plausible-sounding attributions rather than expressing uncertainty, and that citation behavior for the same query was often inconsistent across repeated trials from the same system.

Rao, Wong, and Callison-Burch (2026) provide the most extensive quantitative treatment of agentic citation reliability to date, examining citation URL validity across ten commercial LLMs and deep research agents on 53,090 URLs from the DRBench benchmark and three models on 168,021 URLs from ExpertQA spanning 32 academic fields. They report that 3–13% of citation URLs are outright hallucinated — meaning the URL has no archival record in the Internet Archive's Wayback Machine and most likely never existed — while 5–18% are non-resolving overall once link rot and other retrieval failures are included. A key comparative finding is that deep research agents generate substantially more citations per query than search-augmented chatbots, yet hallucinate URLs at a higher rate per citation, indicating that the apparent thoroughness of agentic outputs does not translate into proportionally higher citation reliability. Domain effects were also pronounced, with non-resolving rates ranging from 5.4% in business-related queries to 11.4% in theology-related queries.

Finally, large-scale archival and cross-model benchmarking work has begun to quantify fabrication at the level of the published scholarly record itself rather than only at the level of individual model outputs. Xu

et al. (2026), in the GhostCite study, benchmarked 13 state-of-the-art LLMs on citation generation across 40 research domains and additionally analyzed 2.2 million citations drawn from 56,381 papers published at top-tier AI/ML and security venues between 2020 and 2025, finding that all 13 models hallucinated citations at rates between 14.23% and 94.93% depending on domain, and that 1.07% of the sampled published papers themselves contained invalid or fabricated citations — with an 80.9% year-over-year increase in such cases observed in 2025, a pattern the authors link, cautiously and without asserting direct causation, to the growing use of LLMs in manuscript preparation. Related audits of citation popularity bias, such as Algaba, Mazijn, Holst, Tori, Wenmackers, and Ginis (2024) and the analysis embedded in Naser (2026), further find that when LLMs do cite real papers, they disproportionately favor highly cited, already-prominent works — a secondary distortion of the scholarly record that compounds the primary fabrication problem.

3. Main Thematic and Technical Discussion: A Taxonomy and Comparative Benchmark Landscape

3.1 A Taxonomy of Citation Fabrication

Synthesizing the studies reviewed above, LLM and agentic citation failures can be organized into four overlapping categories. (1) Wholesale fabrication, in which a cited work does not exist under any title, author combination, or identifier — the majority failure mode identified by Walters and Wilder (2023) and replicated across nearly every subsequent study. (2) Metadata corruption, in which a cited work exists but core bibliographic details (volume, issue, page numbers, DOI, or PMID) are incorrect, a category Walters and Wilder found affecting 24–43% of nominally "real" citations depending on model generation. (3) Identity mismatch, in which real author names or real venue names are attached to fabricated titles, or a genuine identifier (a DOI or PMID) is attached to the wrong paper — the pattern first documented by Alkaissi and McFarlane (2023) and formalized as a detection criterion ("substantial author-identity mismatch") in later verification pipelines. (4) Unsupported attribution, most prominent in agentic search and news-citation contexts, in which a real, existing source is cited but does not actually support the claim attributed to it, or a syndicated republication is cited in place of the original source — the central failure mode documented by the Tow Center's news-citation study (Jaźwińska & Chandrasekar, 2025) and distinguished explicitly from pure non-existence by Rao, Wong, and Callison-Burch (2026), who frame attribution failure and existence failure as related but analytically separate questions: "does the source exist?" versus "does the source support the claim?"

3.2 Comparative Fabrication Rates Across Systems

Table 1 summarizes representative quantitative findings drawn from the studies reviewed in Section 2, illustrating both the wide range of reported rates and the systematic factors — model generation, domain, and system architecture — that explain much of this variance.

System / Study	Year	Fabrication / Hallucination Rate	Source
ChatGPT (GPT-3.5), literature reviews	2023	55% of citations fabricated; 43% of real citations erroneous	Walters & Wilder (2023)
ChatGPT (GPT-4), literature reviews	2023	18% of citations fabricated; 24% of real citations erroneous	Walters & Wilder (2023)
ChatGPT (GPT-3.5), natural sciences vs. humanities	2024	27.3% (sciences) vs. 23.4% (humanities) non-existent; DOI hallucination 61.8–89.4%	Mugaanyi et al. (2024)
GPT-4o, mental health literature reviews	2025	19.9% overall; 6% (high-familiarity topic) to 28–29% (low-familiarity topic)	Linardon et al. (2025)
13 commercial LLMs, 40 research domains	2026	14.2%–94.9% across models and domains	Xu et al. (2026), GhostCite
10 commercial LLMs, cross-model citation audit	2026	11.4%–56.8% across models, domains, and prompts	Naser (2026)
Legal LLM queries (case citations)	2024	58%–88% hallucination depending on model and query type	Dahl et al. (2024)
Commercial AI legal research tools (RAG-based)	2025	17%–33% hallucination despite retrieval grounding	Magesh et al. (2025)
8 generative AI search engines, news attribution	2025	60%+ incorrect citations overall (37% best-performing to 94% worst-performing)	Jaźwińska & Chandrasekar (2025)
10 LLMs / deep research agents, citation URLs	2026	3%–13% hallucinated URLs; 5%–18% non-resolving overall; agents higher than search-augmented LLMs	Rao et al. (2026)

Several patterns emerge from this comparison. First, model scale and generation matter: the drop from 55% (GPT-3.5) to 18% (GPT-4) fabrication reported by Walters and Wilder (2023) is the most frequently cited evidence that later-generation models are meaningfully, though not completely, more reliable citers. Second, retrieval augmentation reduces but does not eliminate fabrication: commercial RAG-based legal research tools evaluated by Magesh et al. (2025) still hallucinated in 17–33% of outputs despite explicit vendor claims of hallucination-free operation, a finding consistent with the theoretical point made by Lewis et al. (2020) that retrieval augmentation improves factual grounding by conditioning generation on retrieved passages but does not guarantee that the generator will faithfully represent those passages rather than reverting to parametric generation when retrieval is incomplete or ambiguous. Third, agentic and search-integrated systems, despite having live web access, are not uniformly more reliable than closed-book chatbots — the Tow Center's finding that generative search engines were wrong more than 60% of the time on direct-quotation attribution tasks, and Rao et al.'s (2026) finding that deep research agents hallucinate URLs at higher per-citation rates than search-augmented LLMs despite issuing more

citations overall, together indicate that adding retrieval or agentic browsing capability shifts the character of the fabrication problem (from invented bibliographic metadata to invented or non-resolving URLs and misattributed news sources) without resolving it. Fourth, domain and topic familiarity are consistently significant moderators, with fabrication rates rising for less prominent topics, non-English or humanities sources, and lower-income-country contexts (Linardon et al., 2025; Mugaanyi et al., 2024).

3.3 Why Citation Presence Itself Can Be Destabilizing

Recent benchmark work has extended the question beyond how often models fabricate citations to ask how the mere presence of a citation — real or fabricated — affects a model's downstream epistemic behavior. Khurana, Ramana RN, and Kumar (2026), in the AuthorityBench study, used a fully balanced 2×2 factorial design crossing claim veracity (true or false) with citation veracity (real or fabricated) across general knowledge, science, law, and medicine, evaluating seven models on 220,564 prompts. They found that citation presence — whether the citation was real or fabricated — increased hallucination rates relative to a no-citation baseline in every model tested, and that this effect was strongest when a fabricated citation was attached to an otherwise true claim, in some cases pushing model agreement with an unrelated or incorrect elaboration up by 3 to 22 percentage points, and as high as 35–77% in the general-knowledge domain specifically. This finding has direct implications for agentic search products: because such systems routinely attach citations to their outputs as a credibility signal, and because users are shown to treat the presence of a citation as evidence of reliability regardless of whether that citation is genuine, citation-augmented generation can, under some conditions, make models more rather than less prone to confidently asserting unsupported content.

4. Current Approaches and Developments in Detection and Mitigation

4.1 Retrieval-Augmented Generation

The dominant architectural mitigation strategy is retrieval-augmented generation (RAG), introduced by Lewis et al. (2020), in which a generator model conditions its output on passages retrieved from an external, non-parametric knowledge source rather than relying solely on knowledge encoded in model weights. In principle, RAG should sharply reduce citation fabrication because the model is asked to summarize or cite specific retrieved documents rather than to recall bibliographic metadata from memory. In practice, the legal-domain evaluation by Magesh et al. (2025) demonstrates that RAG substantially reduces but does not eliminate hallucination, because retrieval can fail silently (returning no relevant passage, which the generator then fills in from parametric memory), because generators can still misattribute or distort retrieved content, and because multi-step legal or research queries often require synthesizing across several retrieved documents in ways that reintroduce fabrication at the synthesis stage.

4.2 Sampling-Based and Black-Box Detection

Where model internals or retrieval provenance are unavailable — as is the case for most commercial chatbots and agentic search products accessed only through a user interface or API — detection has relied on black-box, sampling-based methods. Manakul, Liusie, and Gales (2023) introduced SelfCheckGPT, a

zero-resource approach based on the principle that if a model genuinely "knows" a fact, independently sampled responses to the same prompt will tend to agree, whereas hallucinated content will vary and contradict itself across samples; the method requires no external database and has been adapted in several citation-specific verification pipelines as a first-pass filter. A related, citation-specific variant of this consistency logic is the multi-model consensus filter proposed by Naser (2026), who found that citations independently generated by more than three of ten tested LLMs achieved 95.6% verified accuracy — a 5.8-fold improvement over single-model citation reliability — alongside a within-prompt repetition filter based on the same underlying intuition that spurious, fabricated citations are less likely to recur consistently than genuine ones.

4.3 Post-Hoc Verification Pipelines

A growing body of applied tooling focuses on post-hoc, database-grounded verification rather than trying to prevent fabrication at generation time. Xu et al. (2026) developed CiteVerifier, an open-source framework that combines robust reference parsing with a cascaded, multi-source retrieval strategy across local bibliographic indexes, academic databases (such as CrossRef, OpenAlex, and Semantic Scholar), and web search, using similarity-based matching to classify each citation as valid or invalid at scale; the same underlying strategy of cross-referencing multiple authoritative bibliographic sources appears in Naser's (2026) verification of 69,557 citation instances against CrossRef, OpenAlex, and Semantic Scholar simultaneously. In the URL-citation setting specific to agentic search and deep research tools, Rao, Wong, and Callison-Burch (2026) proposed and evaluated a detection-and-correction pipeline that cross-checks cited URLs against the Internet Archive's Wayback Machine to distinguish genuinely non-existent (hallucinated) links from merely non-resolving (link-rotted) ones, and explored whether post-hoc verification could be used not only to flag but to actively correct citations before they reach the end user.

5. Research Challenges and Limitations

Despite the volume of empirical work reviewed above, several methodological and structural limitations complicate direct comparison and slow the field's progress toward standardized, actionable benchmarks.

- **Definitional inconsistency:** studies vary in how strictly they define "fabrication," with some counting only wholly non-existent works and others also counting substantial metadata errors, author-identity mismatches, or unsupported attribution as failures, making raw fabrication-rate percentages difficult to compare across studies without careful attention to methodology (compare, for example, the narrower existence-only criterion used by Rao et al. (2026) with the broader multi-category criteria used by Walters & Wilder (2023)).
- **Small and non-representative samples in early work:** several of the foundational case studies and cross-disciplinary comparisons rely on fewer than 700 total citations across a handful of prompted topics (Walters & Wilder, 2023; Mugaanyi et al., 2024), raising questions about how well percentage estimates generalize across the much larger space of possible prompts, domains, and phrasing that real-world users employ.

- **Verification labor and scalability:** manual verification against library databases, as used in the earliest studies, does not scale to the hundreds of thousands or millions of citations examined in more recent large-scale audits, forcing a shift toward automated, database-matching verification pipelines whose own false-positive and false-negative rates are not always independently validated.
- **Rapid model turnover and lack of longitudinal tracking:** because commercial LLMs and agentic products are updated frequently and often without public versioning or changelogs for retrieval and prompting infrastructure, benchmark results reported at one point in time may not remain valid even a few months later, and few studies re-test the same systems longitudinally to establish whether fabrication rates are systematically improving, stagnant, or fluctuating with unannounced product changes.
- **Opacity of commercial agentic systems:** deep research agents and generative search engines are closed systems whose retrieval sources, ranking, and citation-generation logic are not publicly documented, meaning that benchmark studies can characterize output-level fabrication rates but cannot fully diagnose whether a given failure originates in retrieval, ranking, synthesis, or citation formatting — a limitation acknowledged explicitly by both the Tow Center study (Jaźwińska & Chandrasekar, 2025) and Rao et al. (2026).
- **Domain and language skew:** the large majority of existing benchmarks are conducted in English and concentrate on biomedical, legal, and computer-science domains, leaving citation reliability in the humanities, non-English scholarship, and lower-resource languages comparatively under-studied even though the limited existing evidence (Mugaanyi et al., 2024) suggests these contexts may exhibit higher fabrication rates.

6. Research Gaps and Future Directions

Building on the challenges identified above, we propose several priorities for future benchmarking and mitigation research.

- **Standardized, multi-category fabrication metrics:** the field would benefit from a shared taxonomy and scoring rubric — extending early efforts such as Aljamaan et al.'s (2024) reference hallucination score — that consistently distinguishes wholesale fabrication, metadata corruption, identity mismatch, and unsupported attribution, enabling apples-to-apples comparison across studies and systems.
- **Continuous, longitudinal benchmarking infrastructure:** rather than one-off snapshots, the field needs recurring, versioned evaluations (analogous to model leaderboards in other NLP subfields) that track fabrication rates for the same set of commercial LLMs, RAG products, and agentic search tools over time as underlying models and retrieval infrastructure change.
- **Benchmarks that jointly test existence and support:** as Rao et al. (2026) note, existing work has focused disproportionately on whether a cited source exists rather than on whether it actually supports the claim attributed to it; future benchmarks should systematically evaluate both dimensions together, particularly for agentic search tools whose primary failure mode in the Tow

Center study was misattribution to real but wrong or non-supportive sources rather than pure non-existence.

- Open, reproducible verification infrastructure for publishers and courts: tools such as CiteVerifier (Xu et al., 2026) and RefChecker-style pipelines demonstrate technical feasibility for automated pre-publication citation auditing; integrating such tools directly into manuscript-submission systems, legal e-filing platforms, and journalistic fact-checking workflows would convert reactive detection into preventive infrastructure, addressing the finding that fabricated citations are already entering the peer-reviewed and legal record undetected.
- Causal and mechanistic understanding of fabrication: most existing work is descriptive (measuring how often fabrication occurs) rather than mechanistic (explaining why); further work connecting citation fabrication to training-data density, decoding strategy, and calibration research such as SelfCheckGPT's consistency-based framework (Manakul et al., 2023) could inform training-time interventions rather than purely post-hoc filtering.
- Equity-focused evaluation: given preliminary evidence that fabrication rates are elevated for humanities sources, non-English contexts, and lower-income-country references (Mugaanyi et al., 2024), dedicated benchmarks assessing whether citation unreliability disproportionately disadvantages already under-represented scholarship are needed to prevent AI-assisted research tools from further marginalizing this work.
- User-facing calibration and interface design: because AuthorityBench-style findings (Khurana et al., 2026) show that citation presence itself can increase user and model trust independent of citation validity, future work should evaluate interface-level interventions — confidence indicators, inline verification badges, or mandatory link-checking before display — as a complement to backend detection methods.

7. Conclusion

Citation fabrication has moved from an anecdotal curiosity noted in early ChatGPT case studies to a rigorously, if unevenly, benchmarked property of large language models and the agentic systems built upon them. The empirical record assembled across dozens of studies since 2023 shows clear, if incomplete, progress: fabrication rates have fallen substantially from GPT-3.5-era levels near 55% to GPT-4-class levels near 18–20%, and retrieval-augmented systems further reduce, though do not eliminate, the problem, typically landing in the 15–35% range even in commercial products explicitly engineered and marketed to avoid hallucination. At the same time, the newest and most autonomous category of tool — agentic deep research systems and generative search engines that browse and cite the live web — has not resolved the problem and in some respects has introduced new failure modes, including high rates of misattribution to real-but-wrong sources and elevated per-citation hallucination despite producing more citations overall. As these tools become further embedded in scholarly publishing, legal practice, and everyday information-seeking, the research community's priority should shift from repeatedly re-discovering that fabrication occurs toward building the standardized metrics, longitudinal benchmarks, and verification infrastructure needed to monitor, disclose, and ultimately reduce citation fabrication as an ongoing property of a rapidly evolving technology landscape rather than as a static, one-time finding.

References

- Algaba, A., Mazijn, C., Holst, V., Tori, F., Wenmackers, S., & Ginis, V. (2024). Large language models reflect human citation patterns with a heightened citation bias. *arXiv*. <https://arxiv.org/abs/2405.15739>
- Alkaissi, H., & McFarlane, S. I. (2023). Artificial hallucinations in ChatGPT: Implications in scientific writing. *Cureus*, 15(2), e35179. <https://doi.org/10.7759/cureus.35179> (PMID: 36811129)
- Aljamaan, F., Temsah, M.-H., Altamimi, I., Al-Eyadhy, A., Jamal, A., Alhasan, K., Mesallam, T. A., Farahat, M., & Malki, K. H. (2024). Reference hallucination score for medical artificial intelligence chatbots: Development and usability study. *JMIR Medical Informatics*, 12, e54345. <https://doi.org/10.2196/54345>
- Chen, S., Yin, H., & Zhou, S. (2026). LegalCiteBench: Evaluating citation reliability in legal language models. *arXiv:2605.10186*. <https://arxiv.org/abs/2605.10186>
- Dahl, M., Magesh, V., Suzgun, M., & Ho, D. E. (2024). Large legal fictions: Profiling legal hallucinations in large language models. *Journal of Legal Analysis*, 16(1), 64–93. <https://doi.org/10.1093/jla/laae003> (arXiv:2401.01301)
- Jażwińska, K., & Chandrasekar, A. (2025, March 6). AI search has a citation problem. *Columbia Journalism Review*, Tow Center for Digital Journalism. https://www.cjr.org/tow_center/we-compared-eight-ai-search-engines-theyre-all-bad-at-citing-news.php
- Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), Article 248. <https://doi.org/10.1145/3571730> (arXiv:2202.03629)
- Khurana, A., Ramana RN, A., & Kumar, D. (2026). Authority, truth, and citation bias: A large-scale multi-domain benchmark for studying epistemic susceptibility in large language models. *arXiv:2606.13104*. <https://arxiv.org/abs/2606.13104>
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan, & H. Lin (Eds.), *Advances in Neural Information Processing Systems* 33 (pp. 9459–9474). Curran Associates. (arXiv:2005.11401)
- Linardon, J., Jarman, H. K., McClure, Z., Anderson, C., Liu, C., & Messer, M. (2025). Influence of topic familiarity and prompt specificity on citation fabrication in mental health research using large language models: Experimental study. *JMIR Mental Health*, 12, e80371. <https://doi.org/10.2196/80371>
- Magesh, V., Surani, F., Dahl, M., Suzgun, M., Manning, C. D., & Ho, D. E. (2025). Hallucination-free? Assessing the reliability of leading AI legal research tools. *Journal of Empirical Legal Studies*, 22(2), 216–242. <https://doi.org/10.1111/jels.12413> (arXiv:2405.20362)
- Manakul, P., Liusie, A., & Gales, M. J. F. (2023). SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (pp. 9004–9017). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.emnlp-main.557> (arXiv:2303.08896)
- Mugaanyi, J., Cai, L., Cheng, S., Lu, C., & Huang, J. (2024). Evaluation of large language model performance and reliability for citations and references in scholarly writing: Cross-disciplinary study. *Journal of Medical Internet Research*, 26, e52935. <https://doi.org/10.2196/52935>

- Naser, M. Z. (2026). How LLMs cite and why it matters: A cross-model audit of reference fabrication in AI-assisted academic writing and methods to detect phantom citations. arXiv:2603.03299. <https://arxiv.org/abs/2603.03299>
- Rao, D., Wong, E., & Callison-Burch, C. (2026). Detecting and correcting reference hallucinations in commercial LLMs and deep research agents. arXiv:2604.03173. <https://arxiv.org/abs/2604.03173>
- Walters, W. H., & Wilder, E. I. (2023). Fabrication and errors in the bibliographic citations generated by ChatGPT. *Scientific Reports*, 13, 14045. <https://doi.org/10.1038/s41598-023-41032-5> (PMID: 37679503; PMCID: PMC10484980)
- Xu, Z., Qiu, Y., Sun, L., Miao, F., Wu, F., Wang, X., Li, X., Lu, H., Zhang, Z., Hu, Y., Li, J., Luo, J., Zhang, F., Luo, R., Liu, X., Li, Y., & Liu, J. (2026). GhostCite: A large-scale analysis of citation validity in the age of large language models. arXiv:2602.06718. <https://arxiv.org/abs/2602.06718>